

HW3

1) ~~$x[n]u[n]$~~

1) ~~$x[n]u[5-n]$~~

$x[n]$	$x[n]u[5-n] = y[n]$
$x[-2] = 0$	$x[-2]u[5-(-2)] = 0$
$x[-1] = 0$	$x[-1]u[5-(-1)] = 0$
$x[0] = 1$	$x[0]u[5-0] = 1$
$x[1] = 2$	$x[1]u[5-1] = 2$
$x[2] = 1$	$x[2]u[5-2] = 1$
$x[3] = 2$	$x[3]u[5-3] = 2$
$x[4] = 2$	$x[4]u[5-4] = 2$
$x[5] = 0$	$x[5]u[5-5] = 0$
$x[6] = 1$	$x[6]u[5-6] = 0$
$x[7] = 0$	$x[7]u[5-7] = 0$

2) $y[n] = x[5-n](u[n+2] - u[n-5])$

$y[n]$

-2	$x[5-(-2)](u[2+2] - u[-2-5]) = 0$
-1	$x[5-(-1)](u[-1+2] - u[-1-5]) = 1$
0	$x[5](u[2] - u[-5]) = 0$
1	$x[4](u[3] - u[-4]) = 2$
2	$x[3](u[4] - u[-3]) = 2$
3	$x[2](u[5] - u[-2]) = 1$
4	$x[1](u[6] - u[-1]) = 2$
5	$x[0](u[7] - u[0]) = 0$
6	$x[-1](u[8] - u[1]) = 0$
7	$x[-2](u[9] - u[2]) = 0$

2) $h[n]$

1)

$$x[n] = 2\delta[n-1] + 2\delta[n-2] + 2\delta[n-3] - \delta[n-4] + \delta[n-5]$$

2) $h[n]$

n	-2	-1	0	1	2	3	4	5	6	7	8	9	
$x[n]$	0	0	0	2	-2	2	-1	1	0				
$x[n]$	0	1	2	3	2	1	0	0	0				
	0	0	0	2	-2	2	-1	1	0				
		0	0	0	4	-4	2	-2	2	0			
			0	0	0	6	-6	6	-3	3	0		
				0	0	0	4	-4	4	-2	2	0	
					0	0	0	2	-2	2	-1	1	0
$y[n]$	0	0	0	2	2	4	1	3	1	3	1	1	0

3) Causal, $h[n] = 0$ when $n < 0$